

DOCUMIND

By:Janani V R

Project Title: DocuMind: AI-Powered Intelligent Document Analysis

System

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Subject: GenerativeAI

Introduction

With the exponential growth of digital documents, extracting meaningful insights efficiently has become a significant challenge. Traditional document readers lack interactivity and intelligence, making information retrieval time-consuming.

DocuMind addresses this issue by combining **Artificial Intelligence, Natural Language Processing, and Web Development** to create an intelligent document assistant.

The platform allows users to:

- Upload and manage documents
- Interact with documents through conversational AI
- Generate quizzes and flashcards
- Visualize knowledge through mind maps
- Evaluate AI responses for quality assurance

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Abstract

DocuMind is an AI-powered web application designed to transform how users interact with documents. The system enables users to upload PDF files and perform intelligent operations such as querying content, generating quizzes, extracting highlights, and visualizing concepts.

By integrating modern web technologies with advanced Large Language Models (LLaMA 3 via Groq), DocuMind provides a seamless and interactive platform for learning, research, and knowledge extraction.

The system utilizes a Retrieval-Augmented Generation (RAG) pipeline to ensure accurate and context-aware responses. Additionally, it incorporates evaluation mechanisms to measure AI response quality, making it a robust and production-ready intelligent system.

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Objectives

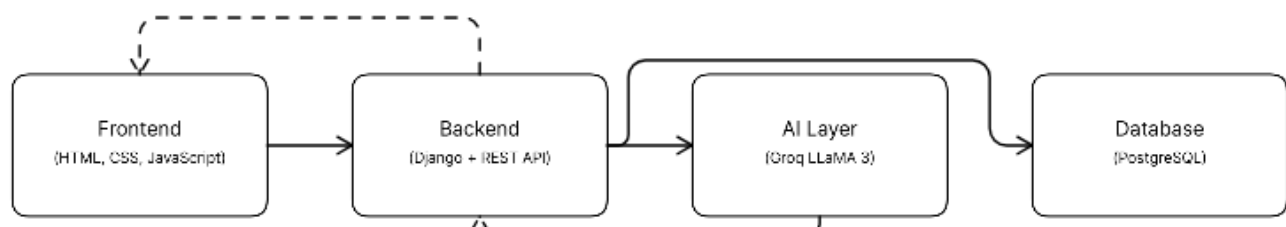
The main objectives of DocuMind are:

- To build a secure user authentication system
- To enable PDF upload and intelligent text extraction
- To implement a RAG-based chat system
- To generate quizzes and flashcards automatically
- To evaluate AI-generated responses using defined metrics
- To highlight key concepts intelligently
- To visualize document knowledge through mind maps

SystemArchitecture

- The system follows a layered client-server architecture integrating frontend, backend, database, and AI components.
- The frontend (HTML, CSS, JavaScript) provides the user interface and communicates with the backend via APIs.
- The backend (Django + REST Framework) handles business logic, authentication, and API management.
- The database (PostgreSQL) stores user data, documents, chat history, and generated content.
- The AI layer (Groq LLaMA 3) performs intelligent processing like answering queries, generating quizzes, and extracting insights.
- The system uses a Retrieval-Augmented Generation (RAG) approach for context-aware responses.
- Data flows from user actions → backend processing → AI layer → response back to frontend.
- The architecture ensures scalability, modularity, and efficient performance.

System Architecture



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Modules and Features

8.1 User Authentication Module

Purpose:

To ensure secure access and manage user identity within the system.

Functionality:

This module enables users to register, log in, and log out securely using JWT-based authentication. It protects restricted routes by verifying tokens and ensures that only authorized users can access system features. Additionally, it manages user profile details such as avatar, bio, and account information, providing a personalized experience.

Status: Completed

8.2 Document Management Module

Purpose:

To handle the uploading, processing, and storage of PDF documents.

Functionality:

Users can upload PDF files, which are then processed to extract textual content using PyPDF2. The extracted text is divided into smaller chunks for efficient retrieval and analysis. The module stores document metadata such as title, upload date, and page count, enabling organized document management and quick access.

Status: Completed

8.3 Chat with PDF (RAG Module)

Purpose:

To enable intelligent interaction with documents through natural language queries.

Functionality:

This module allows users to ask questions related to uploaded documents. It retrieves the most relevant text chunks using similarity search and sends them to the LLaMA 3 model to generate context-aware responses. The system also provides citations, indicating the source pages of the answer, ensuring transparency and reliability.

Status: Completed

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8.4 Quiz Generator Module

Purpose:

To automate the creation of quizzes for better learning and assessment.

Functionality:

The module generates multiple-choice questions (MCQs) based on document content using AI. Each quiz includes options, correct answers, and explanations. Users can interact with quizzes through a dynamic interface, track their scores, and review their performance.

Status: Completed

8.5 LLM Evaluation Dashboard

Purpose:

To assess and improve the quality of AI-generated responses.

Functionality:

This module evaluates each AI response using seven key metrics: Relevance, Accuracy, Completeness, Clarity, Citation Quality, Conciseness, and Helpfulness. Scores are displayed through visual elements such as radar charts, along with a history of previous evaluations. This helps in monitoring system performance and ensuring reliable outputs.

Status: Completed

8.6 Smart Highlights Module

Purpose:

To identify and emphasize important information within documents.

Functionality:

The system automatically extracts key phrases from documents and categorizes them into types such as definitions, key concepts, and important facts. These highlights are displayed using color-coded markers for better readability. Users can also export summarized highlights for quick revision.

Status: Completed

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8.7 Flashcard Generator Module

Purpose:

To enhance learning through interactive study tools.

Functionality:

This module generates flashcards consisting of questions and answers derived from document content. It includes an interactive interface with flip-card animations and navigation controls. A study mode allows users to test their knowledge by revealing answers on demand.

Status: Completed

8.8 Concept Mind Map Module

Purpose:

To visually represent relationships between key concepts in a document.

Functionality:

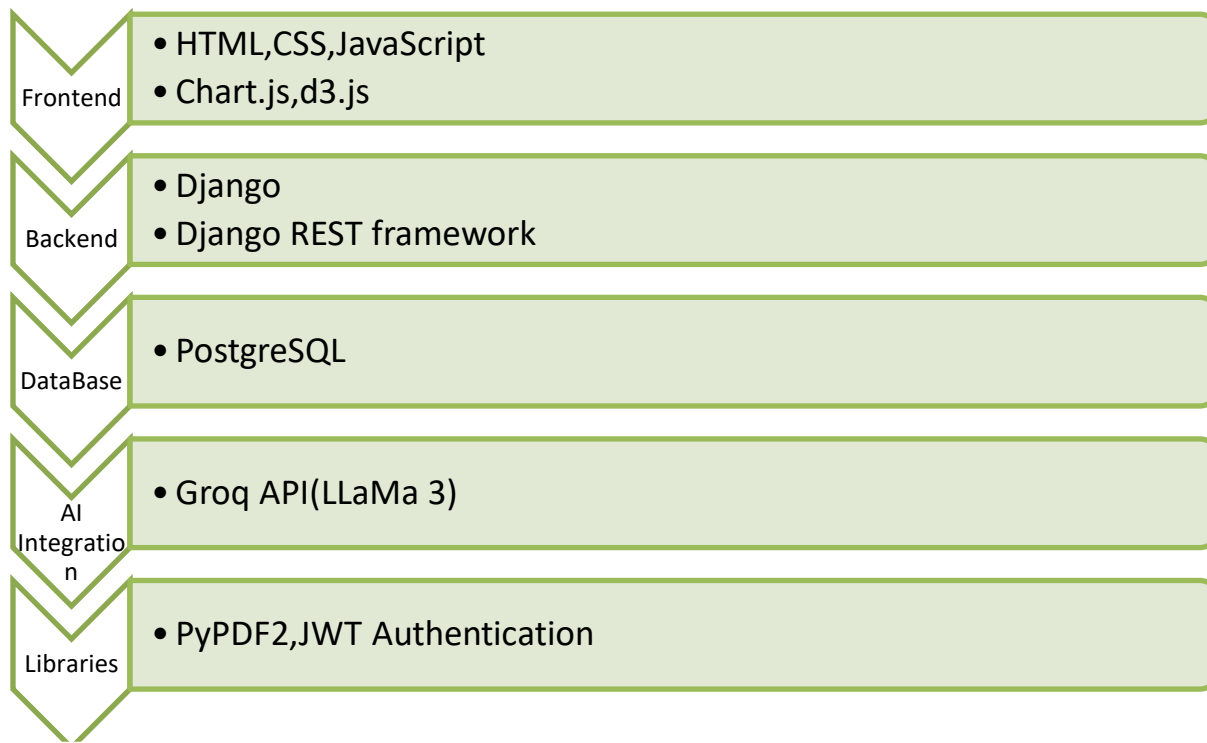
The module extracts main topics and subtopics from the document and organizes them into a structured node graph. Users can interact with the visualization, explore connections, and view detailed descriptions of each concept, improving understanding and retention.

Status: Completed

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Technology Stack



Working Methodology

The working of DocuMind begins when a user uploads a PDF document through the system interface. The backend processes the file by extracting the text and dividing it into smaller chunks, which are stored in the database for efficient retrieval.

When the user asks a question, the system identifies the most relevant text chunks using similarity search. These chunks are combined with the query and sent to the AI model (LLaMA 3), which generates a context-aware response along with source citations.

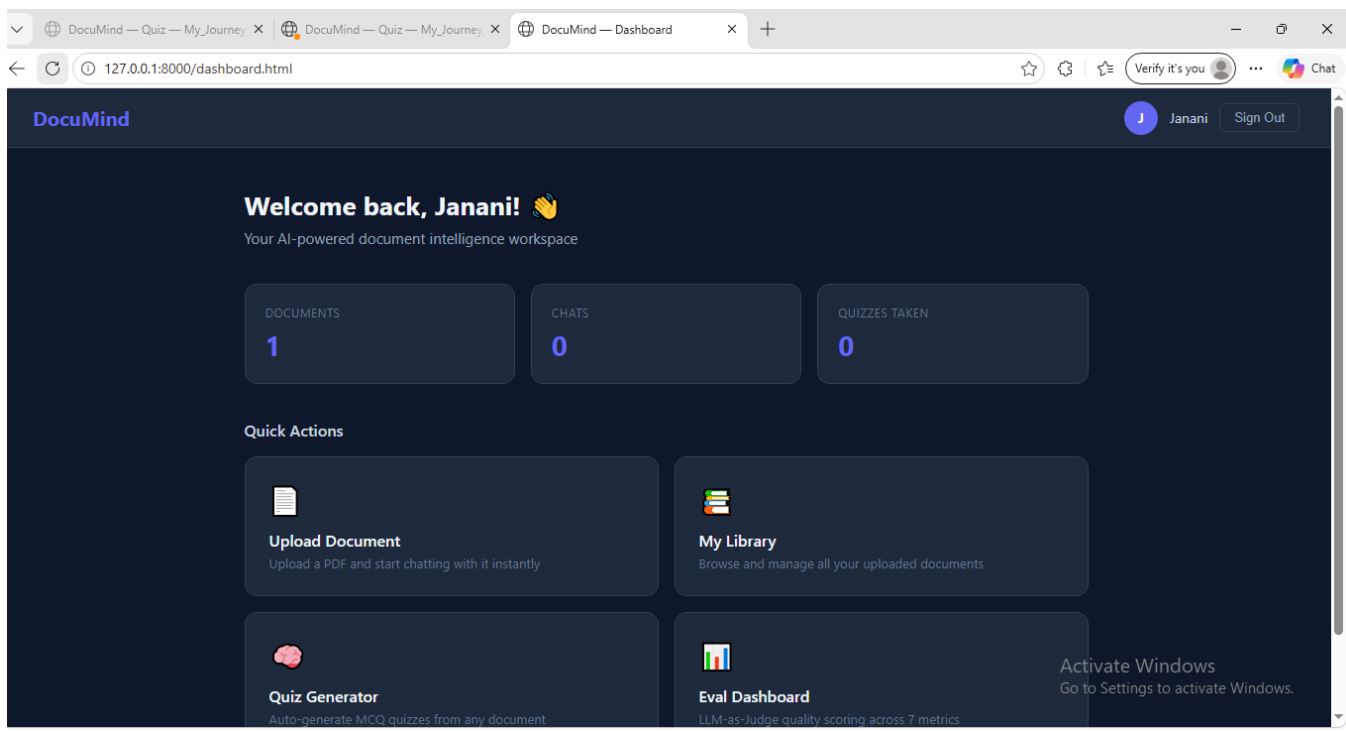
The generated response is then evaluated based on quality metrics such as relevance and accuracy, and the results are stored for analysis. Additionally, the system can generate quizzes, flashcards, and highlights from the document, enhancing the overall learning and interaction experience.

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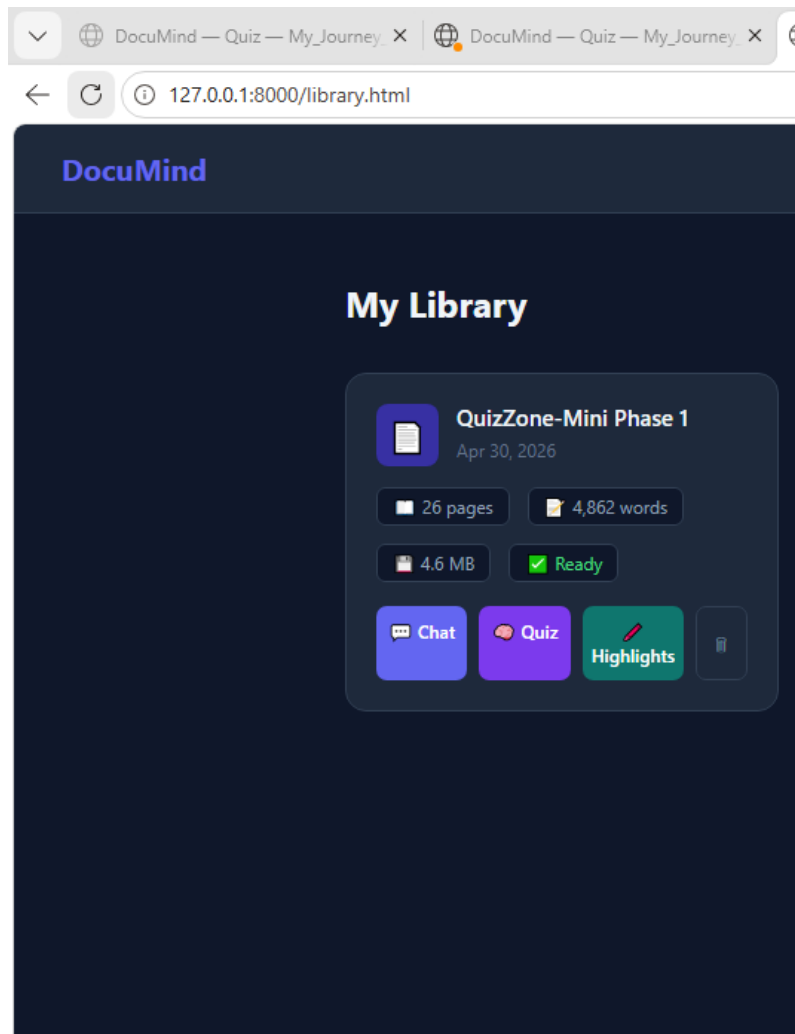
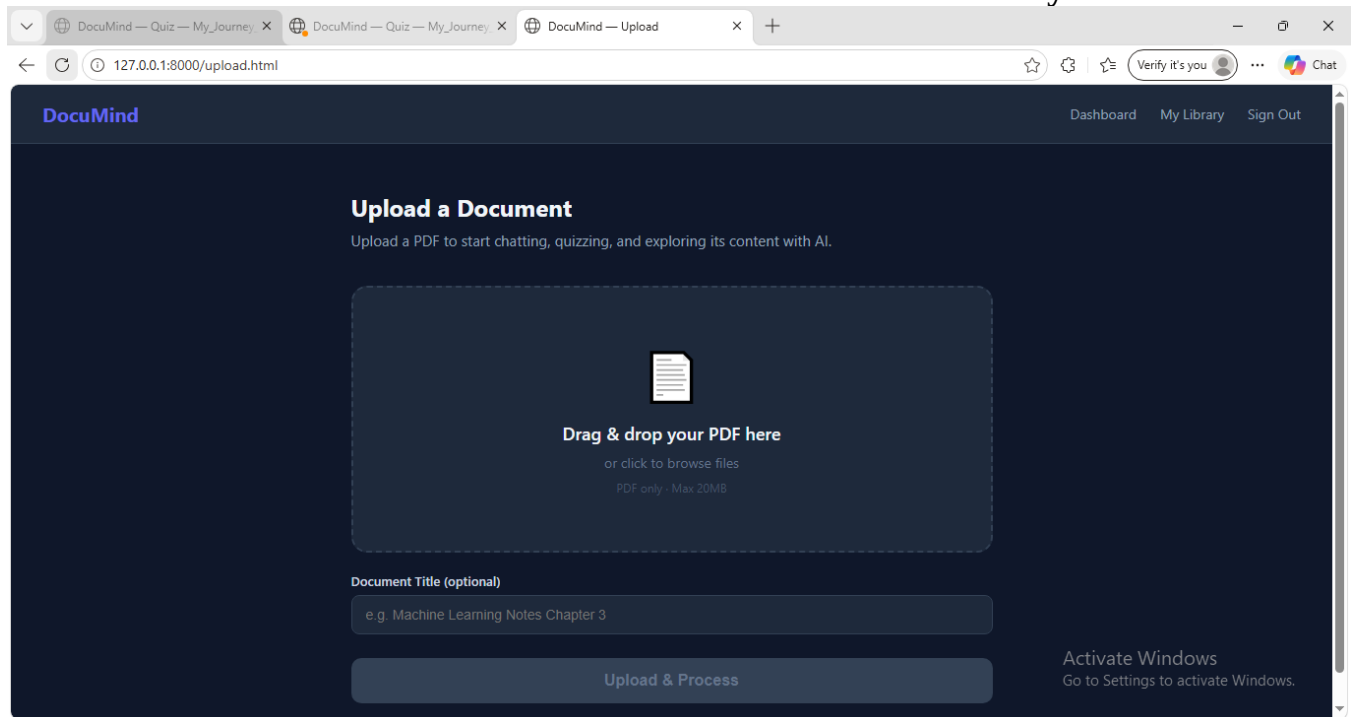
Results Obtained

- 1.Efficient document understanding:**The system enables users to quickly grasp the key ideas of a document through AI-driven summaries, highlights, and contextual responses, reducing the need for full manual reading.
- 2.Reduced time for information retrieval:**By using intelligent search and RAG-based querying, users can directly obtain precise answers instead of scanning entire documents, significantly saving time.
- 3.Automated learning tools (quiz, flashcards):**DocuMind automatically generates quizzes and flashcards from document content, helping users reinforce their understanding without additional effort.
- 4.Improved user engagement:**Interactive features such as chat, quizzes, highlights, and mind maps make the learning process more engaging and user-friendly compared to traditional static systems.
- 5.AI response quality measurement:**The system evaluates AI-generated answers using multiple metrics, allowing users to assess reliability and ensuring consistent improvement in response quality.



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DocuMind QuizZone-Mini Phase 1

DOCUMENT INFO

QuizZone-Mini Phase 1

Pages 26
Words 4,862
Status Ready

TIPS

- Ask specific questions for better answers
- Answers include page citations
- Ask follow-up questions naturally
- Try "Summarize this document"

Ask me anything about this document

I'll find the relevant passages and give you cited answers with page numbers.

Summarize this document What are the main topics? What are the key conclusions? List the most important points

Ask a question about this document...

Activate Windows Go to Settings to activate Windows.

DocuMind QuizZone-Mini Phase 1

Quiz — QuizZone-Mini Phase 1

1 / 5

QUESTION 1 OF 5

What is the main goal of QuizZone?

A To provide a basic question-and-answer functionality

B To offer a personalized and adaptive quiz experience

C To serve as a competitive leaderboard

D To assist users with navigation and platform queries

Previous Next

Activate Windows Go to Settings to activate Windows.

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The screenshot shows a web browser with three tabs: 'DocuMind — Quiz — My_Journey', 'DocuMind — Quiz — My_Journey', and 'DocuMind — Quiz — QuizZone-M'. The address bar shows '127.0.0.1:8000/quiz.html?doc=2'. The DocuMind interface has a dark theme. At the top right are 'Library' and 'Dashboard' buttons. The main content area displays a large '60%' score, '3 out of 5 correct', and a 'Keep studying!' button. Below this are buttons for 'Retake Quiz', 'New Quiz', and 'Back to Library'. An 'Answer Review' section shows a question: 'Q1. What is the main goal of QuizZone?' with a correct answer: 'Your answer: B. To offer a personalized and adaptive quiz experience'. A note below states: 'QuizZone aims to address the gap in existing online quiz platforms by providing personality-driven feedback, real-time'. A Windows watermark is visible on the right side of the interface.

The screenshot shows a web browser with three tabs: 'DocuMind — Quiz — My_Journey', 'DocuMind — Quiz — My_Journey', and 'Highlights — QuizZone-Mini Phase 1'. The address bar shows '127.0.0.1:8000/highlights.html?doc=2'. The DocuMind interface has a dark theme. At the top right are 'Library' and 'Dashboard' buttons. The main content area is titled 'QuizZone-Mini Phase 1' and features a 'Smart Highlights' section with the text: 'Let LLaMA 3 automatically extract and categorize key phrases, concepts, and facts from your document.' Below this is a 'Generating...' button and a status '22 highlights generated.' A summary bar shows: 'Total: 22', 'Key Concepts: 6', 'Facts: 11', and 'Definitions: 1'. An 'All Highlights' section lists two highlights, both categorized as 'KEY CONCEPT'. The first highlight is: '"Online quiz platforms have become essential tools for self-assessment and knowledge building."' (Page 1). The second highlight is: '"QuizZone addresses this gap by integrating personality-driven feedback, real-time countdown timers, a competitive leaderboard, and a rule-based chatbot, assistant."' (Page 1). A sidebar on the left titled 'FILTER BY CATEGORY' lists: 'All Highlights (22)', 'Key Concept (6)', 'Definition (1)', 'Important Fact (11)', 'Methodology (3)', 'Conclusion (1)', and 'Manual (0)'. At the bottom of the sidebar is a section for 'ADD MANUAL HIGHLIGHT' with a text input field, a 'Manual' dropdown, and an 'Add Highlight' button. A Windows watermark is visible on the right side of the interface.

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CONCLUSION

DocuMind successfully demonstrates how modern Artificial Intelligence techniques can be effectively integrated with document processing systems to create a powerful and intelligent platform. By leveraging a **Retrieval-Augmented Generation (RAG) pipeline**, the system is able to provide accurate, context-aware responses based on the content of uploaded documents. This significantly improves the way users interact with large volumes of textual data.

The inclusion of advanced features such as **LLM-based evaluation**, quiz generation, flashcards, smart highlights, and concept visualization enhances both productivity and learning efficiency. Users are not only able to retrieve information quickly but also engage with the content through interactive and automated learning tools. The evaluation mechanism further ensures the reliability and quality of AI-generated responses, making the system more trustworthy and effective.

Overall, DocuMind provides a scalable, modular, and user-friendly solution that addresses the limitations of traditional document systems. It stands as a practical application of AI in real-world scenarios, particularly in education, research, and knowledge management, and has strong potential for further enhancement and deployment.

REFERENCES

1. Django Documentation – Official documentation for backend development and web framework implementation.
- 2.Django REST Framework Documentation – Reference for building RESTful APIs and handling client-server communication.
- 3.Groq API Documentation – Used for integrating LLaMA 3 for AI-based text generation and evaluation.
- 4.PyPDF2 Documentation – Library used for extracting text from PDF documents.
- 5.PostgreSQL Documentation – Reference for database design, queries, and data management.
- 6.Research Papers on Retrieval-Augmented Generation (RAG) – For understanding the methodology behind context-aware AI responses.

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